

What to Consider When Considering the Audience

Martynas Jočys February 1, 2024



Source: Karolina Grabowska // Pexels.com

“Considering the audience” is one of the most important pieces of advice in storytelling and data visualization. It is generally suggested to think about their needs, goals, and ability to understand the items displayed in the visual.

This allows our message to reach them more easily and resonate with them better, but the idea of “considering the audience” itself

is quite vague. Some of us have a natural sense of empathy and “get” the right approach quickly, while some of us don’t and need a more concrete checklist.

For example, we have guidelines on how to deal with colour blindness – which is a trait our audience might have, but examples like these are usually floating alone.

In this article, I will turn the vague suggestion “consider the audience” into something more checklist-like, which you could go through and consider the audience “considered”.

1. Data Literacy

Since data visualization can be described as showing **data**, we can start investigating our audience on how familiar they are with data in general.

Highly data-literate people will likely come from engineering, data science, or data analytics fields. Counterintuitively, some non-technical majors also have strong statistical backgrounds, like psychology. They will be able to understand more sophisticated statistical charts like scatterplots or boxplots, and will have less

trouble grasping more difficult statistical concepts like confidence intervals and linear models.

Less data-literate people might still be highly educated, but their education might come from a different field – like literature or fine arts. They won't be as familiar with more sophisticated charts. For them, simpler things like bars, lines and dots will be a safer option. Also, do not overload them with statistical concepts; instead, explain a Machine Learning model in terms of a friendly robot sipping warm oil.



Source: Martynas Jočys

2. Subject Knowledge

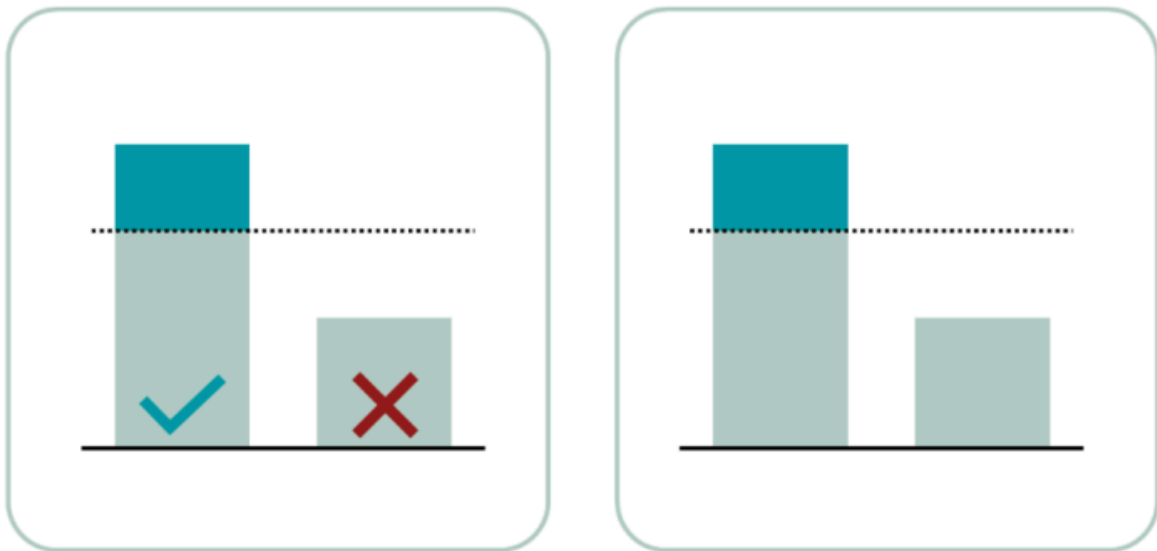
Another thing to consider is whether people know the subject matter very well or if it is something completely new to them.

Subject matter experts will want to skip introductions and explanations. They may be fine with most abbreviations and may even prefer them to full words. Sometimes, there might be some strongly preferred data visualisation conventions, and any deviations might be considered too creative and out of place!

For example, while making a presentation for finance professionals, you don't need to explain what EBITDA or WACC are, or even dismantle the abbreviation. Finance professionals will prefer some specific way the data is displayed, most probably mimicking profit-loss statements. The waterfall chart is spotted showing finance data way more often than any other kind of information. But be careful – finance people might be able to spot tiny inaccuracies quickly!

People who are unfamiliar with the subject matter will need those introductions and explained abbreviations. This means there will be less time to analyze the insight. The insight itself is best to be highlighted because its meaning might not be obvious. (If WACC is going **up**, is it good or bad?). You might need to explain what

“good” and “bad” means, and what the consequences in every case are – this might even become the whole focus of your data visualization.



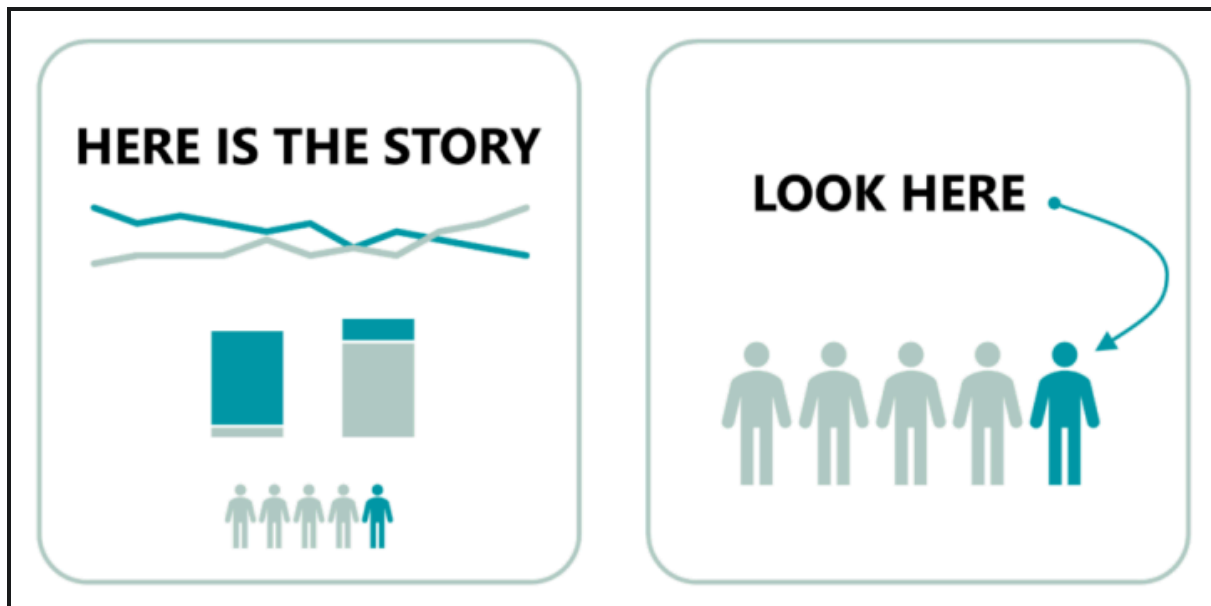
Source: Martynas Jočys

3. Time Span

Time is related not to the audience themselves, but more to the situation they're in. We often hear that C-level managers do not have huge attention spans, but this might be the case for other colleagues as well, especially when they consume a chart while skipping through a business newspaper. However, the same CEO might enjoy a long read of his favourite business magazine after working hours.

When people have little time, we must jump to the “aha moment”, the conclusion, and the actionable points as fast as necessary, but not as fast as **possible!** – if you have 10 minutes and the audience is not familiar with the topic, you still need to introduce them, albeit super quick. There is no time to waste, you need to be sharp, focused, super clear, and to the point.

When people have more time, there is still a difference in what kind of time they have – is it their leisure time, or is the meeting just incredibly long? We will have more details on that later (see section 8), but more time means you can prepare the audience – making sure they understand the topic, setting, and situation. Also, you have the opportunity to use all the best practices of storytelling – building up the anticipation, tension, and emotion – if this is a prepared presentation. You can also give users a lot of context and many interaction possibilities if this is a dashboard or other interactive visual. But still, no one wants to spend time with your chart forever – it still needs to be sharp, focused, super clear, and to the point.



Source: Martynas Jočys

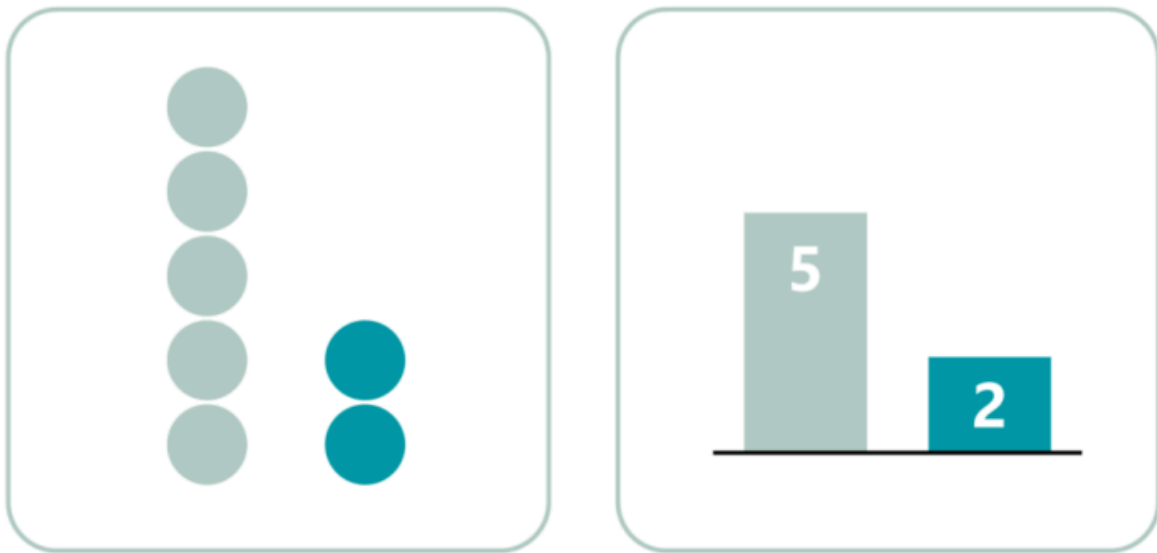
4. There Are Non-Math People

If you're presenting to coworkers, you might know which people simply do not think in terms of math. Some audiences might have more of these people, some consist entirely of them. Think about presenting to children – they might enjoy charts and images, but math is just not their thing unless they're young Hawkings.

Those who are math people will be fine with the fundamentals we do – charts, numbers, averages – all okay.

Those who are non-math people will need something different. Instead of showing numbers, show proportions. Add more annotations and more explanations for them to read. Storytelling

and analogies become much more important for delivering the idea, so add sequences of images and illustrations.



Source: Martynas Jočys

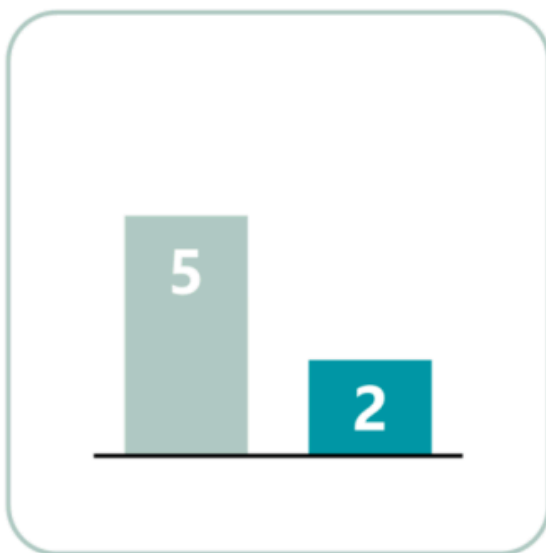
5. There are Non-Chart People

Yes, sadly some people are not that good at decoding charts. They just perceive charts as images to be nice, not something to be read.

Once, a C level manager at my previous employer was reviewing my reports and asked, “The charts are nice, but where can I read about them?” From this experience, I might guess that non-chart people are a bit older – the ones not having exposure to Instagram from birth.

Maybe with more time and effort, you could teach them to read charts, but let's jump into a situation when you have an audience which is lightyears away from math and charts and time is limited – there is **no way** to teach them a different way of thinking.

Unlike non-math people, we can still show numbers because they might even be data literate. However, most probably those folks like to read good old newspapers – writing is the path to their hearts and brains. So, log into ChatGPT and write a great prompt for it to write a well-structured and informative executive summary for them!



5 and 2

Source: Martynas Jočys

6. Specific Sensitivities

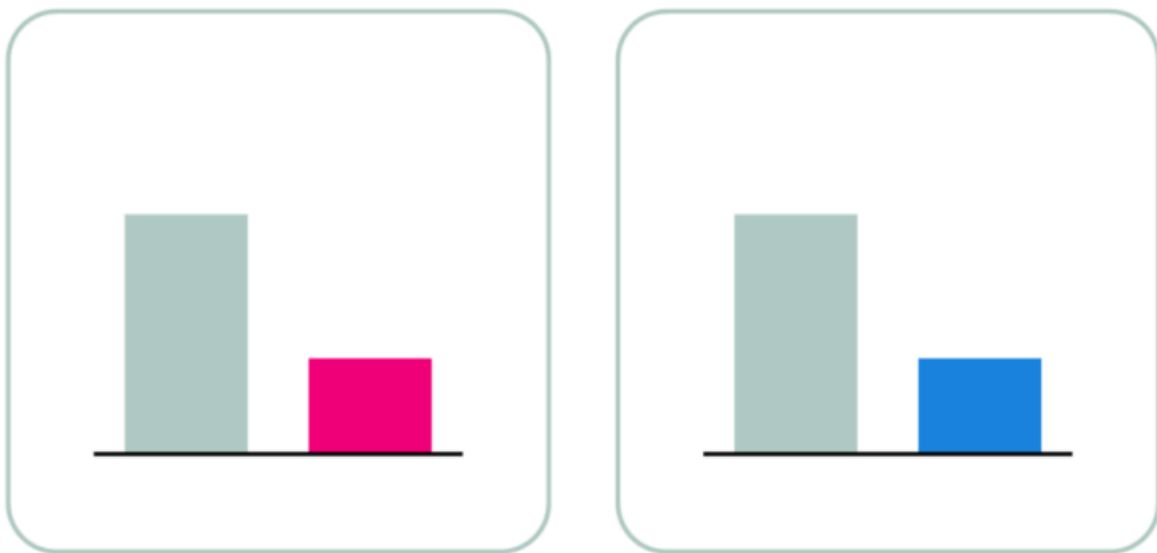
There might be specific sensitivities related to ethnicity, culture, or favourite football team. Neil Richards in his book *Questions in Dataviz* describes a situation when he needed to prepare a presentation about a few British cities. To avoid triggering parts of the audience who love Manchester United, associated with red, and Manchester City, associated with sky blue, he tried to avoid red and blue completely.

Football is fun, and upsetting a fan or two is not the end of the world. But, I'm sure that marking something with the letter Z would negatively trigger audiences from Ukraine, reminding them of their genuine suffering. Meanwhile, mentioning Taiwan in your map wouldn't be very welcomed in a presentation for a Chinese Communist Party, so it should be used according to your political stance and willingness to be kicked out.

Maps are quite a common area of disagreement – even Google Maps notoriously shows different borders of disputed territories depending on whether you're accessing it from China or India.

Such sensitivities can be difficult to figure out. Not being a football fan myself, I would not even consider tiptoeing around the colours of teams; but in the UK, that's a big deal. Research your audience

beyond just their data literacy – get as much context as available, as broad as you have the opportunity – including their culture, history, or even favourite team.



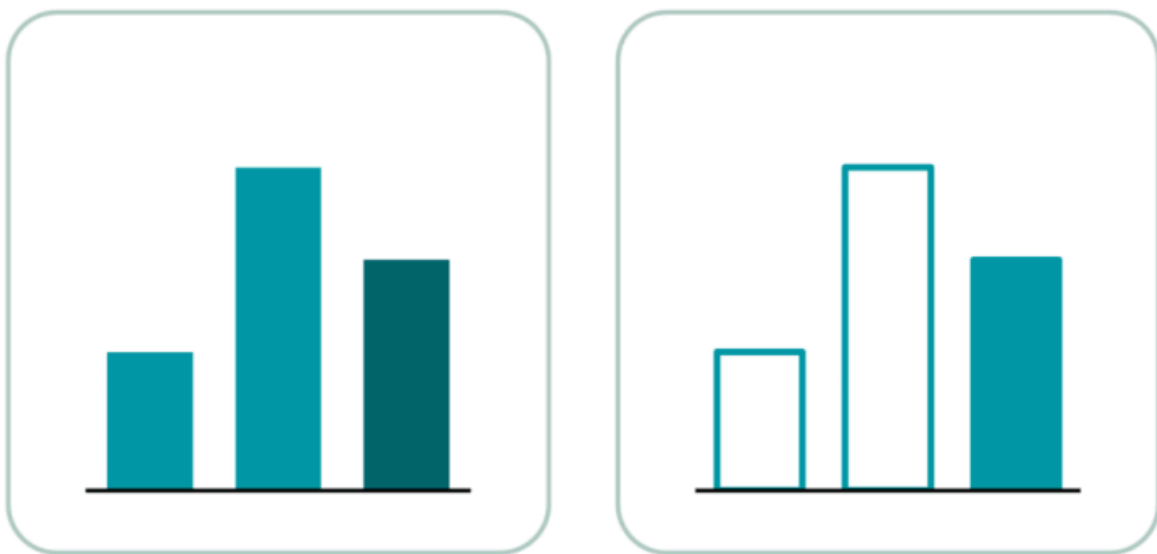
Source: Martynas Jočys

Visual Impairment

I'm not talking only about colour blindness and other types of permanent visual impairment because they can also be situational. Think about poor lighting, when all the light beams are directed towards your projector screen and those vibrant colours in your presentation look bleak. Or about people reading your chart on Uniherz Atom – the smallest smartphone in the market. Or if the whole thing is printed in greyscale.

If you know in advance there will be challenges, you can prepare. Increase contrast, make text bigger, lines thicker, markers larger, shapes stricter and better defined. Do not rely on subtle gradients and don't use them for visual embellishments.

There might be a delicate situation when some audience members are subject to impairment and some aren't, like when the material is printed as handouts, but also presented on screen. In such cases, make sure everyone involved can see what you want to show.



Source: Martynas Jočys

8. Goal

I was talking about football, now I will talk about goals. Goals have the highest impact on the final appearance of the visual. A dashboard for monitoring a process and making immediate decisions vastly differs from a printed infographic made to read, enjoy, and revisit during leisure time. A slide persuading investors about some great opportunity vastly differs from a donut chart in a time-tracking app showing you've spent more time watching YouTube than spreadsheets.

There might be many different ways how to group goals, but this is what I suggest:

- Monitoring a process – a focused chart is needed with a clear indication when something goes off-track.
- Business analysis – someone still needs to convince me that tables are bad. A well-prepared table rich with sparklines, sparkbars, and indicators is your friend.
- Business decision – a summary is needed to compare two or more options side-by-side to highlight which one is better.
- Leisure education – this is where well-prepared infographics and scrollytelling articles shine – people are genuinely curious and have some time, so you can provide them with a lot of details as well as visual cues guiding them towards the main message.
- Leisure just for fun – it could be as artsy as you want, just make sure there are still some hidden information gems the audience could discover!

So, what do you want them to want from your visuals?



Source: Martynas Jočys (left), Stable Diffusion (right)

But What if the Audience is Mixed?

Of course, there are cases when the audience is coming from all sides of the universe, but then the challenge is different. In such a case, you might consider something like a typical Hollywood family movie: when it's generally made for children, but there are spicy bits here and there for adults as well. Did you notice that during *Inside Out* – a nice family movie, there is an on-screen vomiting cat in a post-credit scene?

No matter how mixed the audience is, they still have time and goals. Knowing this, you can adjust your visuals to these factors.

Wrap Up

Did I just make everything more complex? Did a simple “consider your audience” suggestion become an intertwined ball of relations between different balances that might contradict each other? If so, then I did a good job explaining them!

While this all might sound complex, most of the time there are some dominant audience traits we can still define, and I hope that this list will help you spot them quickly. You can always find more nuances, but this depends a lot on how much the audience is already familiar to you, and how much time you have. If it's a bit – you might only consider their subject knowledge or data literacy. If you have more time, find out their most hated colours to avoid.